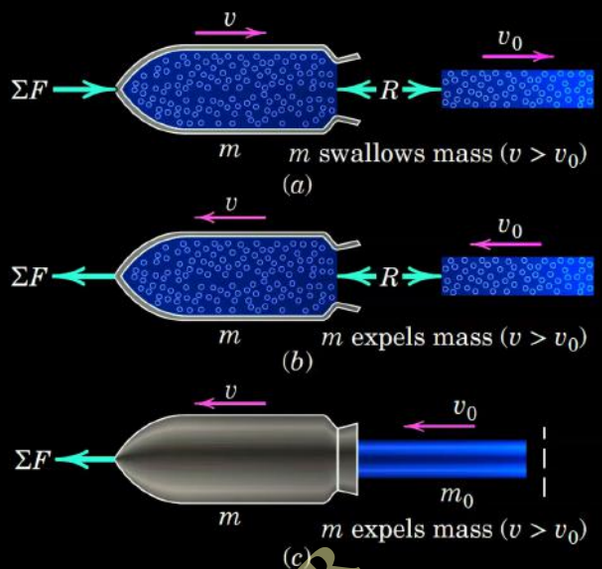
VARIABLE MASS SYSTEMS:



Systems which expel mass or swallow mass are called variable mass systems. During the process of expelling or swallowing the incoming/outgoing mass exerts a force on the bulk mass thus causing its acceleration. Suppose a bulk mass m is moving with a velocity $\vec{v}$ and it swallows a mass dm moving with a velocity $\vec{v}_0$, then due to being absorbed in the bulk mass, the momentum of the incoming mass changes by $dm\,(\vec{v}-\vec{v}_0)$, thus if dt is the time elapsed in the process, the reaction force experienced by this incoming mass is $\vec{F}_R = d\vec{p}/dt = \frac{dm}{dt}(\vec{v}-\vec{v}_0)$, a reaction to this force is experienced by the bulk mass, thus force experience by bulk mass $\vec{F}_{th} = \frac{dm}{dt}(\vec{v}_0-\vec{v}) = \dot{m}\,\vec{v}_{rel}$

where $\vec{v}_{rel}$ is the velocity of incoming mass relative to the bulk mass. Now if $\vec{F}_{ext}$ is the external force on the bulk mass, and $\vec{a}$ is its acceleration, by applying Newton's second law instantaneously, we get:

$$\vec{F}_{ext} + \vec{v}_{rel}\frac{dm}{dt} = m\vec{a} \quad (1)$$ or $$\vec{F}_{ext} + \vec{F}_{th} = m\frac{d\vec{v}}{dt}$$ where $$\vec{F}_{th} = \vec{v}_{rel}\frac{dm}{dt}$$

$\vec{F}_{th}$ = Thrust force of incoming mass.

This is the fundamental equation governing all variable mass systems. Here if mass is being expelled, dm/dt will be taken as negative, $\vec{v}_{rel}$ is always the relative velocity of incoming/outgoing elemental mass wrt bulk mass.

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In case mass is being added / expelled at multiple holes then we need to consider mass flow rates through all routes and accordingly our equation becomes:

$$\boxed{\vec{F}_{ext} + \Sigma \dot{m}_i \vec{v}_{rel,i} = m\vec{a}}$$ where $\dot{m}_i$ is mass flow rate at i^{th} inlet or outlet.

<u>Caution:</u> One may be tempted to write $\vec{F}_{ext} = \frac{d}{dt}(mv) = m\frac{dv}{dt} + v\frac{dm}{dt}$

However this is not correct because it does not anywhere take in to account the velocity of mass being expelled. To see this clearly, let us write equation ① clearly:

$$\vec{F}_{ext} + (\vec{v}_0 - \vec{v})\frac{dm}{dt} = m\frac{d\vec{v}}{dt} \Rightarrow$$

$$\vec{F}_{ext} = m\frac{d\vec{v}}{dt} + \vec{v}\frac{dm}{dt} - \vec{v}_0\frac{dm}{dt}$$, this is not $m\dot{\vec{v}} + \dot{m}\vec{v}$ unless $\vec{v}_0 = 0$.

An alternate way of expressing variable mass equation is

$$\boxed{\vec{F}_{ext} = m\dot{\vec{u}} + \dot{m}\vec{u}}$$ ① where $\vec{u} = \vec{v} - \vec{v}_0$ i.e relative velocity of bulk mass wrt incoming mass.

<u>ALTERNATE DERIVATION:</u> Let m_0 be the mass of a small incoming shaft moving with velocity $\vec{v}_0$. Then,

$$\vec{F}_{ext} = d\vec{p}/dt = \frac{d}{dt}[m\vec{v} + m_0\vec{v}_0] = \dot{m}\vec{v} + m\dot{\vec{v}} + \dot{m}_0\vec{v}_0$$

But $\dot{m}_0 = -\dot{m}$ (∵ mass is conserved), thus $\vec{F}_{ext} = \dot{m}(\vec{v} - \vec{v}_0) + m\dot{\vec{v}}$

$\Rightarrow \vec{F}_{ext} = m\dot{\vec{v}} + \dot{m}\vec{u}$ QED.

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Example: Rocket Propulsion:



The initial combined mass of a rocket and its fuel is m_0. A total mass m_f of fuel is consumed at a constant rate of $dm_e/dt = c$ and expelled at a constant speed of u relative to the rocket. Determine the maximum velocity of the rocket, i.e., at the instant the fuel runs out. Neglect the change in the rocket's weight with altitude and the drag resistance of the air. The rocket is fired vertically from rest.

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Solution: Using variable mass equation:

$\vec{F}_{ext} + \vec{v}_{rel}\, dm/dt = m\vec{a}$ thus: (Taking upward +ve)

$$-(m_0 - ct)g + uc = (m_0 - ct)\, dv/dt$$

Separating the variables and integrating:

$$\int_0^v dv = \int_0^t \left(\frac{uc}{m_0 - ct} - g\right) dt$$

$$\Rightarrow \quad v = u \ln\left(\frac{m_0}{m_0 - ct}\right) - gt \qquad (1)$$

When fuel runs out, $ct = m_f$, and $t = m_f/c$ using this in (1):

$$\boxed{v_{max} = u \ln\left(\frac{m_0}{m_0 - m_f}\right) - \frac{g m_f}{c}} \qquad (2)$$

In a gravity free space we can simply put $\vec{g} = 0$. Thus we see that in such a case, final speed will be independent of mass flow rate. In that case,

$$\boxed{v_{max} = u \ln\left(\frac{m_0}{m_0 - m_f}\right)}$$

Mass Flow Rate For Most Effective Propulsion:

For any rocket we would like to have maximum change in velocity for unit mass of fuel burnt. Now equation ② is:

$\vec{F}_{ext} = m\vec{v} + \dot{m}\vec{u}$, in case of rocket:

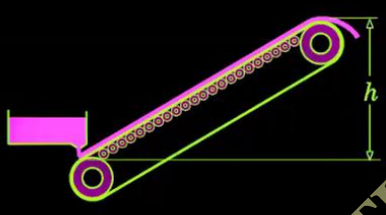
$-mg = m\dot{v} - cu$ $(u = v - v_0)$, c = mass flow rate.

dividing throughout by c:

$\Rightarrow -\frac{mg}{c} = \frac{m\,dv}{dm_f} - u$ where dm_f is mass of fuel burnt in a time dt

$\Rightarrow \frac{dv}{dm_f} = \frac{u}{m} - g/c$ thus to get maximum change in velocity per unit mass of fuel burnt, c should be as large as possible. Thus if there is an option of opening multiple nozzles, all the nozzles should be opened simultaneously.

Example: Pathfinder: WPE: MCQ 21 → 23:



A conveyor belt collects sand and transports it to a height h as shown in the figure. The sand falls on the belt with negligible speed at constant rate μ (mass per unit time). Friction between the belt and the sand particle is so high that the sand particles stop sliding almost instantaneously after they hit the belt. Acceleration due to gravity is g.

21. What should speed of the belt be for the least possible driving force on the belt applied by the motor?

(a) $\sqrt{0.5gh}$ (b) $\sqrt{gh}$

(c) $\sqrt{2gh}$ (d) A speed however small is possible.

22. What is the power delivered by the motor to the belt, when the motor is applying least possible driving force?

(a) $0.5\mu gh$ (b) μgh

(c) $1.5\mu gh$ (d) $2\mu gh$

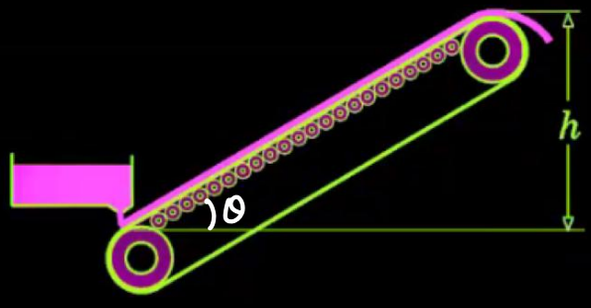
23. What is the power dissipated by the sand-belt system, when the motor is applying least possible driving force?

(a) Zero (b) $0.25\mu gh$

(c) $0.5\mu gh$ (d) μgh

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Solution: 21. Recall that thrust force due to incoming mass is given as, $\vec{F_{th}} = \vec{v_{rel}} \frac{dm}{dt}$. If v is the velocity of conveyor, then this thrust force is μv down the incline. Also if θ is the incline of the conveyor, it carries a mass $\left(\frac{h \sec\theta}{v}\right) \times \mu$



Thus along the incline it experiences a frictional force for supporting the weight of sand, $f = \left(\frac{\mu h \sec\theta}{v}\right) g \sin\theta$ i.e $\mu gh/v$.

Now motors must supply a force to balance the downward f and thrust force. Thus, if F is the force applied by the motors,

$$F = F_{th} + f = \mu v + \mu gh/v, \text{ for } F_{min} \quad dF/dv = 0$$

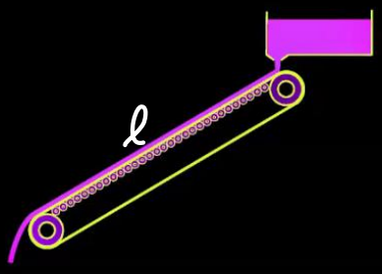
$$\Rightarrow \quad 1 - gh/v^2 = 0 \quad \text{or} \quad v = \sqrt{gh} \text{ m/s (option b)}$$

22.) Now power $P = F_{min} v$, F_{min} @ $v = \sqrt{gh} = 2\mu\sqrt{gh}$, thus least power $P_{min} = 2\mu gh$ Watts (option d).

23.) Power dissipated by kinetic friction pair is frame independent so we might as well calculate it from belt frame. If dm mass falls on the belt in a time dt, its loss in KE, $\frac{1}{2} dm v^2$ thus power developed by friction is $-\frac{1}{2}\mu v^2 = -0.5\mu gh$.
∴ Rate of heat dissipation $Q = 10.5 \mu gh$ (option c).

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EXAMPLE: PATHFINDER: BUILD: 39:

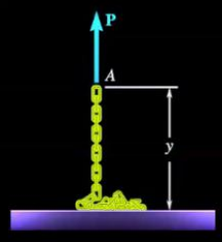


A conveyor belt is inclined at an angle θ with the horizontal. Length of a straight portion of the belt is l, resistive forces in the belt mechanism are negligible and radii of the end cylinders are negligible as compared to the length l. Sand falls on the belt with a negligible speed at a constant rate at the top and leaves at the bottom. Friction between the belt and the sand is high enough to stop sliding of sand particles almost in no time. Inertia of the belt mechanism is negligible as compared to that of the total sand on it. Find the steady speed acquired by the belt. Acceleration due to gravity is g.

Solution: Let μ be the mass flow rate and v be the steady state speed of the conveyor then to maintain constant speed, the thrust force must exactly balance the pulling force of gravity along the incline. Thus $\mu v = \mu\left(\frac{l}{v}\right) g \sin\theta \Rightarrow v = \sqrt{gl\sin\theta}$ Ans.

EXAMPLE (BEER & JOHNSTON):

A chain of length l and mass m lies in a pile on the floor. If its end A is raised vertically at a constant speed v, express in terms of the length y of chain that is off the floor at any given instant (a) the magnitude of the force **P** applied to A, (b) the reaction of the floor.



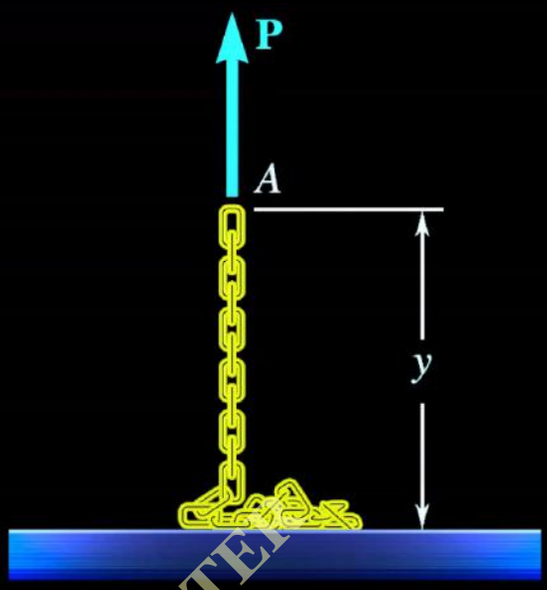
Solution: As the new links are added to the chain, they exert a thrust force, if let λ be mass per unit length then mass part in air is λv and thrust force due to new links being added is then $\vec{F}_{th} = \vec{v}_{rel}\, dm/dt = \lambda v^2$ downwards. Now to maintain a constant speed, P must balance the weight of the chain as well as the thrust force. Thus, $P = \lambda y g + \lambda v^2$

i.e $P = \frac{m}{l}(gy + v^2)$ Ans (a) $(y < l)$

(b) Considering the FBD of the part lying on the floor excluding the link that is just being added to the vertically moving part, normal reaction from the floor must just balance the weight of the chain on the floor. Thus $N = \lambda(l-y)g = \frac{m}{l}(l-y)g$ Ans (b).

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EXAMPLE: Repeat the previous problem if instead of being raised, the chain were being lowered with a constant speed v.



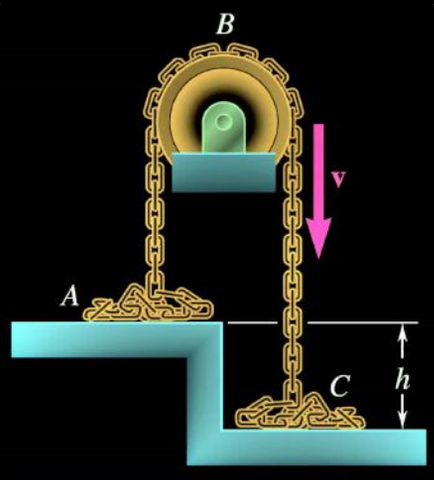
Solution: In this case, instead of links getting tightened at the bottom, they are getting loosened. Thus the tension at the bottom of the vertical part is zero. Thus to lower the chain at constant speed, P must just balance the weight of the vertical part. $\therefore P = \lambda y g = \frac{m}{l} g y$ Ans (a)

Now as the links are forming a heap on the table, the table experiences a thrust force due to collisions with incoming links and in addition, the table has to also support the weight of the chain portion lying on the table. Thus, $N = F_{re} + \frac{m}{l}(l-y)g$

Thus $N = \lambda v^2 + \frac{m}{l}(l-y)g = \frac{m}{l}[v^2 + (l-y)g]$ Ans (b).

EXAMPLE: BEER & JOHNSTON:

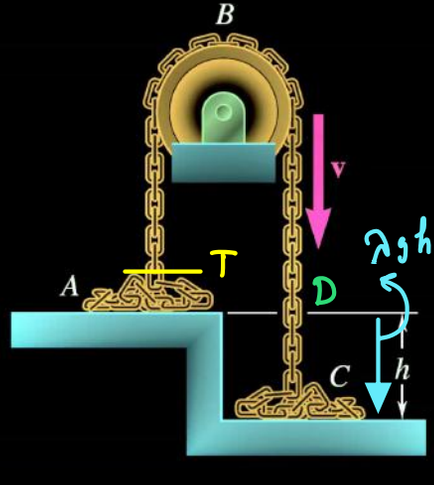
The ends of chain lie in piles at A and C. When released from rest at $t=0$, the chain moves over the pulley B, which has negligible mass. Denoting by L, the length of the chain connecting the two piles and neglecting friction, find the speed v of the chain at a time t.



Solution: Let v be the instantaneous speed of the chain, and λ be linear mass density, then, momentum created on LHS due to rising mass from the pile A in time dt : $dp = (\lambda v dt) v$

$\Rightarrow$ Rate of momentum creation: $T = \lambda v^2$.

Now on RHS there is extra hanging length h creating a pulling force $\lambda g h$.



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Thus net pulling force : $F_p = \lambda g h - \lambda v^2$, Thus:

$$\lambda g h - \lambda v^2 = \lambda L \, dv/dt \quad \Rightarrow \int_0^v \frac{L\,dv}{gh - v^2} = \int_0^t dt$$

$$\Rightarrow \frac{L}{2\sqrt{gh}} \int_0^v \frac{1}{\sqrt{gh} - v} + \frac{1}{\sqrt{gh} + v} \, dv = t$$

$$\Rightarrow \left[-\ln|(v - \sqrt{gh})| + \ln(v + \sqrt{gh}) \right]_0^v = 2\sqrt{gh}\, t/L$$

$$\Rightarrow \text{Until } v < \sqrt{gh}, \quad \ln\left(\frac{\sqrt{gh} + v}{\sqrt{gh} - v}\right) = 2\sqrt{gh}\, t/L$$

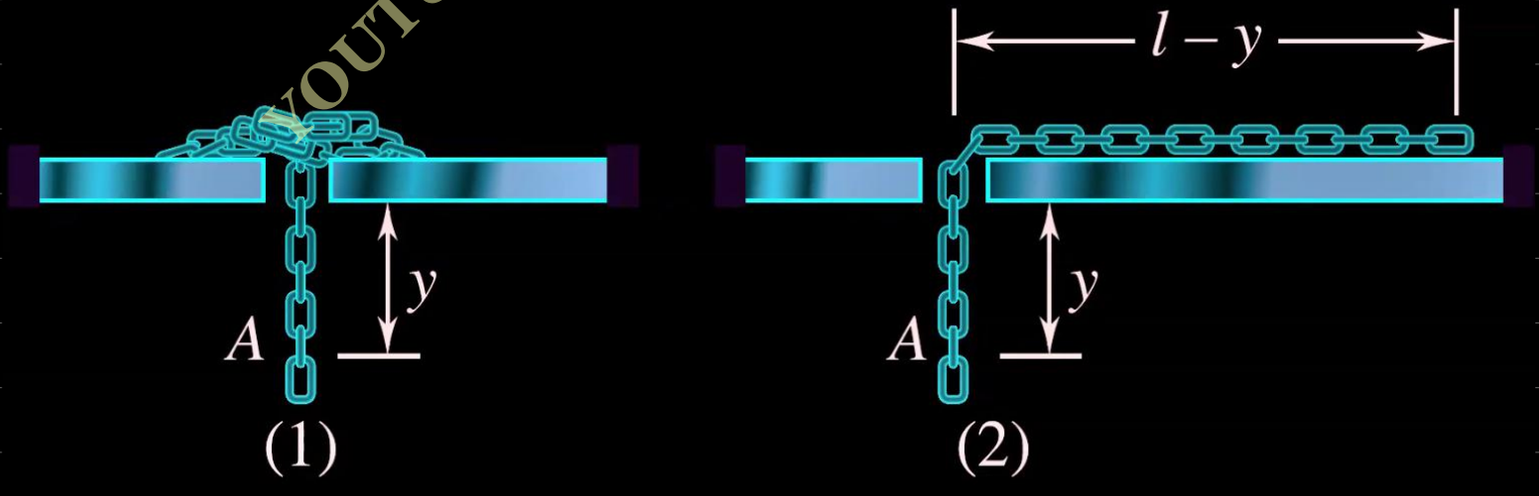
$$\Rightarrow v = \sqrt{gh}\left[\frac{e^{t/\tau} - 1}{e^{t/\tau} + 1}\right] \quad \text{where } \tau = L/2\sqrt{gh}$$

as $t \to \infty$, this attains a steady value of $\sqrt{gh}$.

→ Think: can you find steady speed without integration?

Solution: Steady speed occurs when net pulling force becomes zero thus for steady speed, $\lambda v^2 = \lambda g h \Rightarrow v = \sqrt{gh}$!! ☺.

EXAMPLE: BEER & JOHNSTON:



A chain of length l and mass m falls through a small hole in a plate. Initially when y is very small, the chain is at rest. In each case shown, determine (a) acceleration of the first link as a

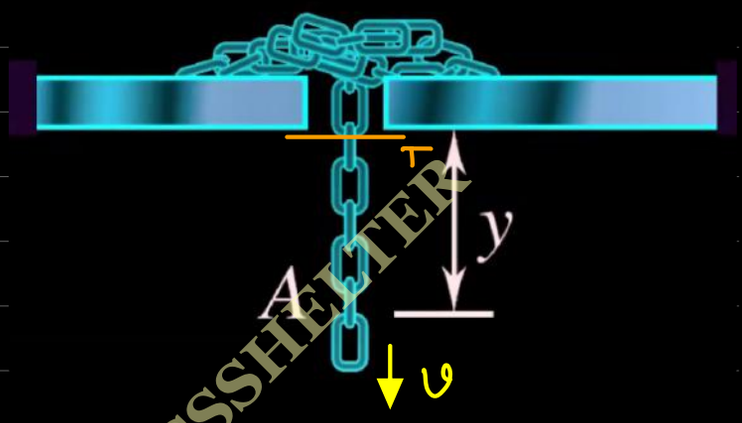
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function of y (b) velocity of the chain as last link passes through the hole. In case 1 assume that individual links are at rest until they fall through the hole. In case 2 assume that at any instant, all links have same speed. Ignore frictional effects.

Solution: (1) Let v be the velocity of the chain at a general y. Then rate of creation of momentum $dp/dt = \lambda v^2$, this must be the tension at the top of the hanging portion. Now applying Second law, $\lambda g y - \lambda v^2 = \lambda a y$

$\Rightarrow v^2 = (g-a)y$, if we want we can solve the differential equation by writing $a = v\,dv/dy$, however recognize that this equation is of the form $v^2 = 2ay$ where a is the constant acceleration, thus by comparing, $2a = g - a$

$\Rightarrow a = g/3$ Ans. Now velocity of last link, $v = \sqrt{2al} = \sqrt{2gl/3}$ Ans.



(2) In this case there are no collisions between the links, therefore, mechanical energy is conserved. Thus:

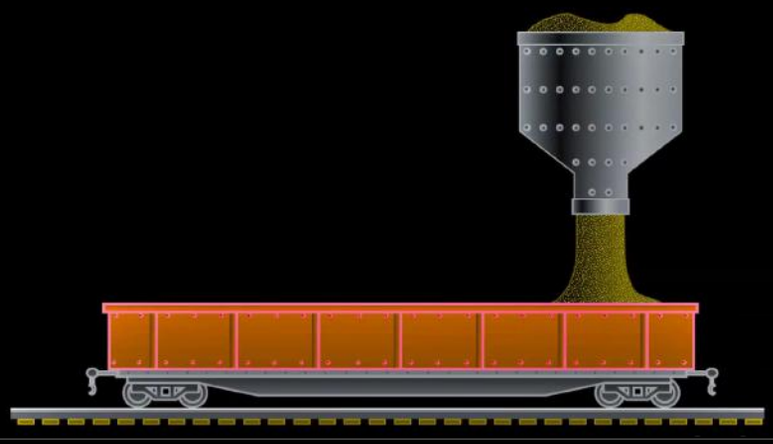
$$\lambda y \times g \times \frac{y}{2} = \frac{1}{2}\lambda l v^2 \qquad \text{(loss in PE = Gain in KE)}$$

Thus, $v = y\sqrt{\frac{g}{l}}$ Ans. Thus when chain's last link leaves,

$v = \sqrt{gl}$ Ans.

EXAMPLE: BEER & JOHNSTON:

A railroad car with length L and mass m_0 when empty is moving freely on a horizontal track while being loaded with sand from a stationary hopper at a rate $dm/dt = q$ Find (a) mass



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of the car by the time it has cleared the hopper (b) Speed of the car by that time.

Solution: Falling sand exerts a thrust force $v\,dm/dt$ in the backward direction where v is the instantaneous speed thus:

$$-vq = (m_0+qt)\,dv/dt \Rightarrow \int_0^t \frac{-q\,dt}{m_0+qt} = \int_{v_0}^{v} \frac{dv}{v}$$

$$\Rightarrow \ln(v/v_0) = -\ln\left(\frac{m_0+qt}{m_0}\right) \Rightarrow v = \left(\frac{m_0 v_0}{m_0+qt}\right) \quad (1)$$

(We could obtain (1) directly also using momentum conservation).

Now time to clear the hopper can be found by integrating velocity:

$\int_0^T v\,dt = L$ where T is time of crossing. Thus:

$$L = m_0 v_0 \int_0^T \frac{1}{m_0+qt}\,dt = \frac{m_0 v_0}{q} \ln\left(\frac{m_0+qT}{m_0}\right)$$

$$\Rightarrow 1+\frac{qT}{m_0} = \exp(qL/m_0v_0) \Rightarrow T = \frac{m_0}{q}\left(e^{qL/m_0v_0}-1\right) \quad (2)$$

now putting (2) in (1), $v = v_0 e^{-qL/m_0v_0}$ Ans.

Mass of the car at this instant: $m = m_0+qT = m_0 e^{qL/m_0v_0}$ Ans.

EXAMPLE: IRODOV: 1.181:

A spaceship of mass m_0 moves in the absence of external forces with a constant velocity $\mathbf{v}_0$. To change the motion direction, a jet engine is switched on. It starts ejecting a gas jet with velocity u which is constant relative to the spaceship and directed at right angles to the spaceship motion. The engine is shut down when the mass of the spaceship decreases to m. Through what angle α did the motion direction of the spaceship deviate due to the jet engine operation?



Solution: Thrust is always sideways which provides centripetal acceleration to the rocket. Thus $uq = mv\omega = mv\,d\theta/dt$

$$\Rightarrow -u\frac{dm}{dt} = mv\frac{d\theta}{dt} \Rightarrow \frac{dm}{m} = -\frac{v_0}{u}d\theta \quad (1)$$

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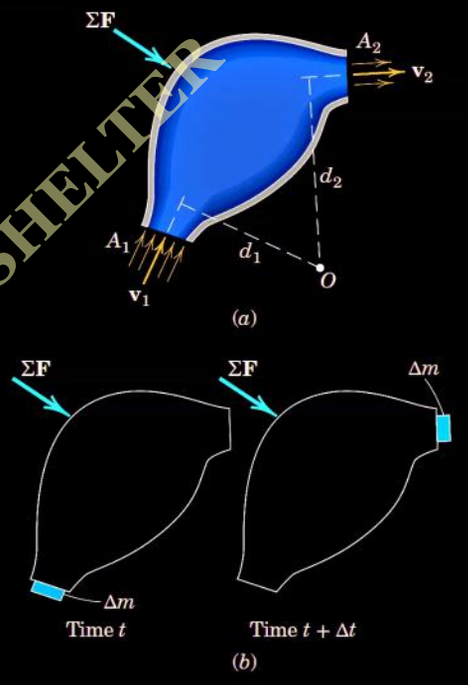
(∵ thrust is always ⊥ to v_0, V_0 remains constant ∴ there is only centripetal acceleration and no tangential acceleration).

Now integrating (1):

$$\int_{m_0}^{m} dm/m = -V_0/u \int_0^{\theta} d\theta$$

$$\Rightarrow \ln(m/m_0) = -\frac{V_0}{u}\theta \Rightarrow \theta = \frac{u}{V_0}\ln(m_0/m) \text{ Ans.}$$

Analysis Of Flow Through A Rigid Container:



Consider a rigid container in which an arbitrary liquid is flowing, such that mass inflow rate is equal to mass outflow rate. The velocity of inflowing stream is $\vec{v}_1$ normal to A_1 and that of leaving stream is $\vec{v}_2$ normal to A_2, then by conservation of mass, $\rho_1 A_1 v_1 = \rho_2 A_2 v_2$. Now if

Let a mass dm enter the container in a time dt then by conservation of mass same mass must also exit the container. Since the flow is steady, velocity at each point in the container remains the same. Thus momentum change of the fluid mass in the container + dm entered mass is simply $dm(\vec{v}_2 - \vec{v}_1)$, thus rate of change of momentum of this much fluid is $d\vec{p}/dt = \frac{dm}{dt}(\vec{v}_2 - \vec{v}_1)$.

This must be the resultant of all the forces acting on the fluid (which includes force due to container + any other force on the stream due to fluid pressure at the inlet or outlet. Thus we can write for the fluid:

$$\boxed{\Sigma \vec{F} = \dot{m}\,\Delta\vec{v}}, \text{ where } \Delta\vec{v} = \vec{v}_2 - \vec{v}_1.$$

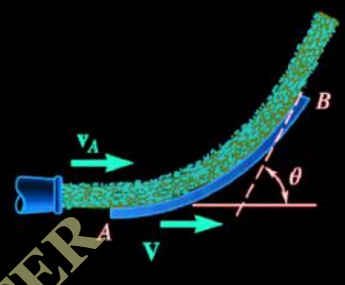
If there are several inlets and outlets, then the

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above equation transforms to: $\Sigma \vec{F} = \Sigma \dot{m}_i \vec{v}_i$ where $\dot{m}_i$ is outflow rate at i^{th} hole (considered negative at inlets) and $\vec{v}_i$ is velocity at i^{th} hole.

EXAMPLE: BEER & JOHNSTON:

A nozzle discharges a stream of water of cross-sectional area A with a velocity $\mathbf{v}_A$. The stream is deflected by a *single* blade that moves to the right with a constant velocity $\mathbf{V}$. Assuming that the water moves along the blade at constant speed, determine (*a*) the components of the force $\mathbf{F}$ exerted by the blade on the stream, (*b*) the velocity $\mathbf{V}$ for which maximum power is developed.



[Density of water is ρ]

Solution: $\Sigma \vec{F}_{ext} = \dot{m} \Delta \vec{v}$, now from frame of the vane, the speed of entering jet is $v_A - V$ which should also be the speed of exiting jet. Thus $\dot{m} = \rho A (v_A - V)$ and the required force then becomes: $\dot{m} \Delta \vec{v}$

$$= \rho A (v_A - V) \times (v_A - V)[\cos\theta\, \hat{i} + \sin\theta\, \hat{j} - 1\, \hat{i}]$$

i.e $\vec{F}$ on water: $\vec{F}_w = \rho A (v_A - V)^2 [(\cos\theta - 1)\hat{i} + \sin\theta\, \hat{j}]$ Ans.

(The individual components are $\hat{i}$ and $\hat{j}$ coefficients). Now, vane experiences a reaction force of the force exerted on water thus $\vec{F}_v = -\vec{F}_w$ and power of this force on the vane is then:

$$\mathcal{P} = \vec{F}_v \cdot \vec{V} = \rho A (v_A - V)^2 V [1 - \cos\theta] \quad (\because \vec{V} = V\hat{i})$$

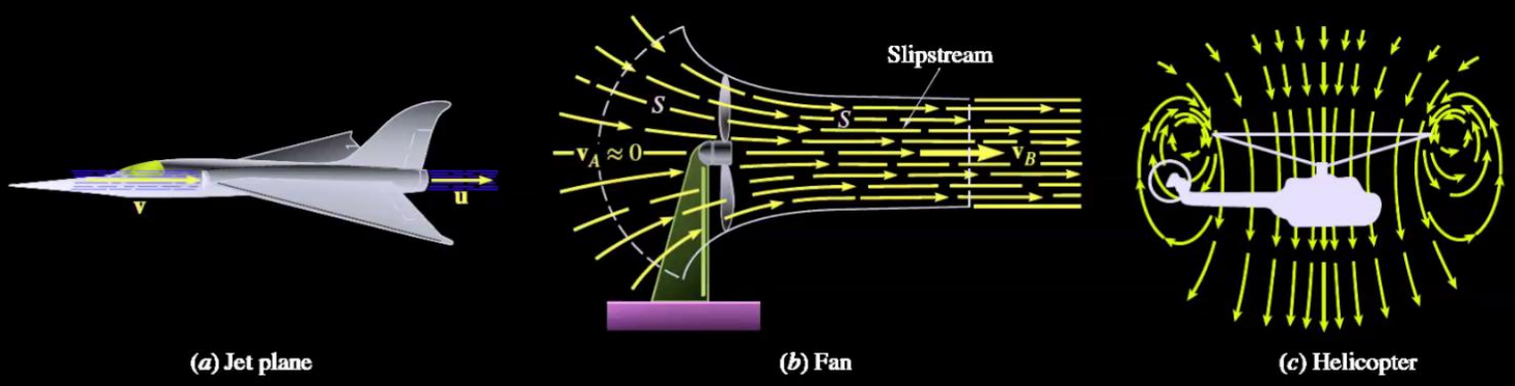
For maximum power, $dP/dV = 0$

$$\Rightarrow V \times 2(v_A - V) \times (-1) + (v_A - V)^2 = 0 \Rightarrow v_A - 3V = 0$$

$$\Rightarrow V = v_A/3 \text{ Ans.}$$

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Jet Engine / Table Fan / Helicopter / Propeller :



(*a*) Jet plane (*b*) Fan (*c*) Helicopter

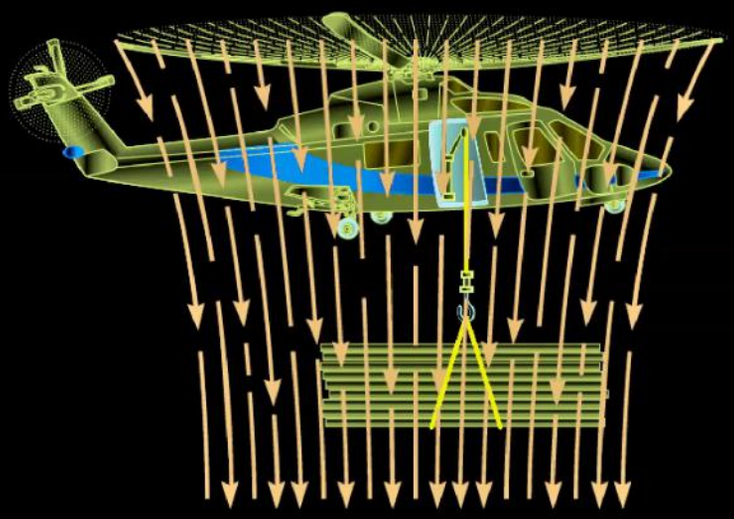
Jet Engine. In a jet engine, air enters the front of the engine with no velocity and leaves through the rear with a high velocity. The energy required to accelerate the air particles is obtained by burning fuel. The mass of the burned fuel in the exhaust gases is usually small enough compared with the mass of the air flowing through the engine that it can be neglected. Thus, the analysis of a jet engine reduces to that of an airstream. We can consider this stream as a steady stream if we measure all velocities with respect to the airplane. We assume, therefore, that the airstream enters the engine with a velocity $\mathbf{v}$ of magnitude equal to the speed of the airplane and leaves with a velocity $\mathbf{u}$ equal to the relative velocity of the exhaust gases (Fig. *a*). Because the intake and exhaust pressures are nearly atmospheric, the only external force we need to consider is the force exerted by the engine on the airstream. This force is equal and opposite to the thrust.

Fan. Consider the system of particles S shown in Fig. *b*. We assume the velocity $\mathbf{v}_A$ of the particles entering the system is equal to zero, and the velocity $\mathbf{v}_B$ of the particles leaving the system is the velocity of the *slipstream*. We can obtain the rate of flow by multiplying v_B by the cross-sectional area of the slipstream. Because the pressure all around S is atmospheric, the only external force acting on S is the thrust of the fan.

Helicopter. Determining the thrust created by the rotating blades of a hovering helicopter is similar to the determination of the thrust of a fan (Fig. *c*). We assume the velocity $\mathbf{v}_A$ of the air particles as they approach the blades is zero, and we obtain the rate of flow by multiplying the magnitude of the velocity $\mathbf{v}_B$ of the slipstream by its cross-sectional area.

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Example: Pathfinder: Build: 40:



Fan of a remote operated model helicopter of mass m has a diameter d. Denoting the density of air by ρ acceleration of free fall by g, deduce an expression for the minimum power required for the take off.

Solution: Force on air stream:

$$F = \rho \frac{\pi d^2}{4} v \,[v - 0] \quad (\because \dot{m} \propto \vec{v}).$$

$$= \frac{\rho \pi d^2 v^2}{4}$$, this must be equal to weight of the chopper.

Thus $$mg = \rho \frac{\pi d^2}{4} v^2 \Rightarrow v = \left(\frac{4mg}{\rho \pi d^2}\right)^{1/2}$$

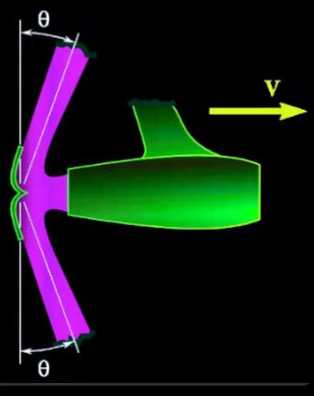
Now Power must also be equal to rate of generation of kinetic energy of the air stream. Thus $P = \frac{1}{2} \dot{m} v^2$

i.e $$P = \frac{1}{2} \rho A v^3 = \frac{1}{2} \rho \cdot \frac{\pi d^2}{4} \left(\frac{4mg}{\rho \pi d^2}\right)^{3/2}$$

$$\Rightarrow P = \frac{(mg)^{3/2}}{d\sqrt{2\rho\pi}}$$ Ans

Example:

In order to shorten the distance required for landing, a jet airplane is equipped with movable vanes that partially reverse the direction of the air discharged by each of its engines. Each engine scoops in the air at a rate of q kg/s and discharges it with a velocity of u m/s relative to the engine. At an instant when the speed of the airplane is V m/s, determine the reverse thrust provided by each of the engines.



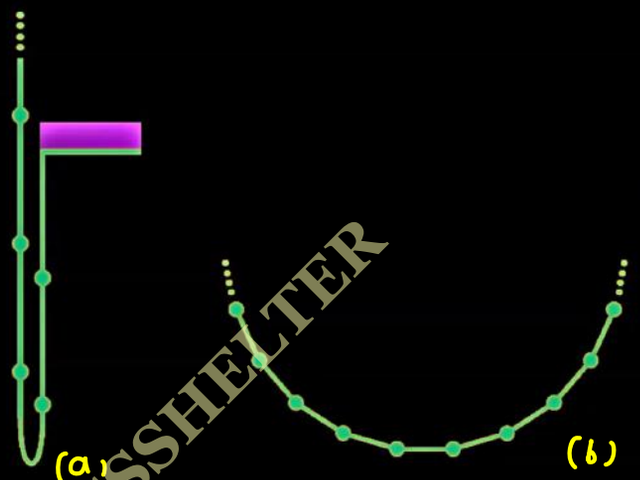
Solution: Here the engine is changing the horizontal momentum of the air jet. From the engine frame, inlet velocity is $-V\hat{i}$ (surrounding atmosphere assumed still).

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Thus thrust on air, $F = \sum \dot{m}_i \cdot \Delta \vec{v}_i = q(u\sin\theta + v)\hat{i}$
This must also be the reverse thrust on the engine.

Energy Conserving And Non Energy Conserving Chains:

A chain may be modeled as a string of beads tied together with massless strings. Suppose a chain has a bend somewhere. Now there can be two scenarios (a) The size of the bend is very small as compared to the spacing between the beads (see first figure) or (b) The loop is sufficiently large as compared to spacing between the beads (second figure). In case (a), The tension in the loop remains zero, but intermittently becomes impulsive as beads from LHS suddenly come to rest due to the constraint of string lengths between beads. This sudden stopping causes loss of energy (which might appear as microscopic deformations / random vibrations and ultimately gets lost to the atmosphere). In the case (b), the particles are not coming to rest all of a sudden, but their speed decreases gradually through the loops as they pass from LHS to RHS due to the chain falling. Thus, there is no sudden jerk and energy loss is avoided. In this chain, the tension at the loop is not zero, infact there the acceleration of LHS is more than g (while in case (a) it was g). The tension at the bottom of vertical part of LHS is responsible for providing the acceleration greater than g. As the moving mass keeps decreasing, the energy gets concentrated in smaller and smaller regions and finally there may appear a noise like that of whip cracking (you might have done this trick with a wet Handkerchief) or in case of chain, the chain will be swaying left to right finally thus loss in gravitational PE will appear as oscillating energy in the swaying of the chain side to side.

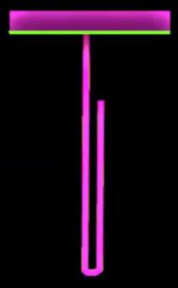


(In energy conserving chain, Tension is almost same on both sides of the bend (assumed small))

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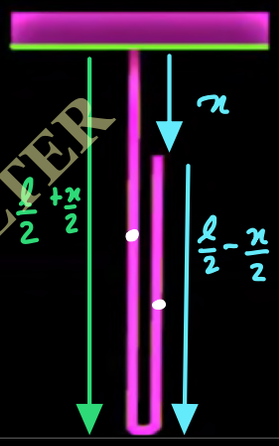
Example: Pathfinder: Build 38:

Ends of a thin uniform inextensible rope of mass $m = 1.0$ kg and length $l = 2.0$ m are fastened to the ceiling close to each other. If one of the ends is released and allowed to fall freely as shown in the figure, find the largest force that the fastening at the other end must be capable of bearing so that rope remains fastened there. Acceleration of free fall is $g = 10$ m/s^2. [Assume non-energy Conserving Rope]



Solution: Let λ be the mass per unit length and consider a general moment when right end of the rope has fallen through a distance x. Now, in the non-energy conserving rope, tension at the bottom RHS point is zero. Velocity of the moving part:

$v = \sqrt{2gx}$ (∵ free fall), Now consider a time dt, the length brought to rest in time dt is $vdt/2$. Thus momentum arrested in this time, $dp = \frac{\lambda v^2 dt}{2}$



on $dp/dt = \frac{\lambda v^2}{2} = \lambda g x$, this is due to tension at bottom most point on LHS. Thus $T_{bottom} = \lambda g x \Rightarrow T_{top} = \lambda\left(\frac{l+x}{2}\right)g + \lambda g x$

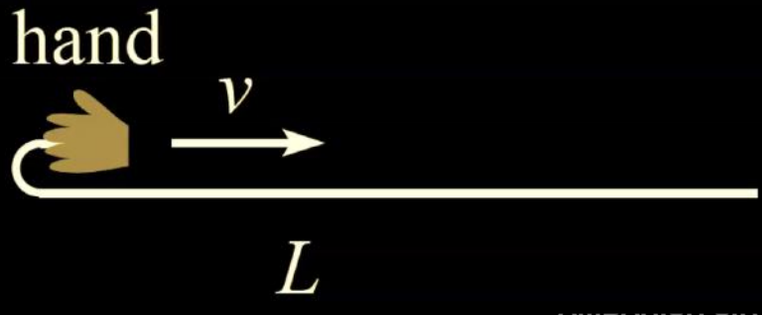
this is maximum when $x = l$, thus, $T_{max} = 2\lambda g l = 2mg$ Ans.

→ Since it is a non-conserving case, the word "chain" would have been more appropriate.

→ With energy conserving chain, with this modeling the momentum would go to zero (∵ $KE = p^2/2m_0$ and $m_0 \to 0$ as $x \to l$. where m_0 is mass of moving portion).

Example: Morin: 5.26:

A chain with length L and mass density λ kg/m lies straight on a frictionless horizontal surface. You grab one end and pull it back along itself in a parallel manner.



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Assume that you pull it at a constant speed v. What force must you apply? What is the total work done by you by the moment chain is straightened out? How much energy is lost as heat if any?

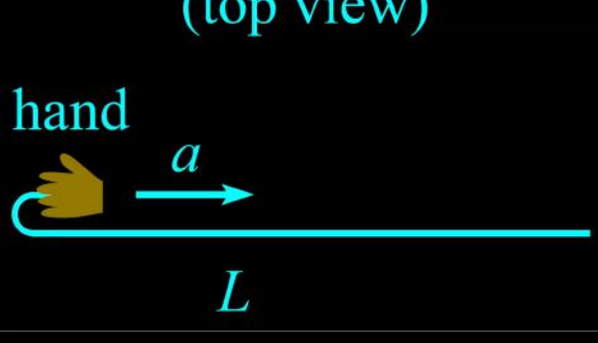
Solution: Mass added to the moving portion in time dt : $dm = \frac{\lambda v dt}{2}$
Thus rate of momentum creation, $dp/dt = \lambda v^2/2$. This must be the tension at the left end of the moving part and also the force exerted by the hand. The hand moves a distance $2L$ by the moment chain is straightened out. Thus work done by hand:
$W = \frac{\lambda v^2}{2} \times 2L = \lambda v^2 L$ Ans.

Now kinetic energy gained by the chain : $\Delta KE = \frac{1}{2}\lambda L v^2$, thus heat liberated : $\Delta Q = \Delta W - \Delta KE = \frac{1}{2}\lambda L v^2$ Ans.
50% of work done is lost as heat.

ALTERNATE METHOD: From frame of hand, chain has initial KE $\frac{1}{2}\lambda L v^2$ and finally it becomes zero. Thus heat lost is $\frac{1}{2}\lambda L v^2$ (from any frame). Now final KE from ground frame is $\frac{1}{2}\lambda L v^2$, thus work done by the hand must be equal to sum of heat liberated and ΔKE i.e
$\frac{1}{2}\lambda L v^2 + \frac{1}{2}\lambda L v^2 = \lambda L v^2$ Ans.

Think: What is the velocity of CM of the moving portion:
Solution: When the right end of the upper part shifts by x, the loop shifts by $x/2$ and CM is the mid point of the two always thus CM shifts by $3x/4$. Thus velocity of CM must be $3v/4$. (Note that momentum of the moving part is not $m_0 v_{cm}$ where m_0 is mass of the moving part, it is actually $m_0 v$).

EXAMPLE: DAVID MORIN:
Suppose in the situation of previous problem you started pulling the chain with constant acceleration instead of constant velocity. What is your force on the chain just before it straightens out.



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Solution: Looking from hand's frame, this is similar to the falling chain problem. The velocity just before straightening out is $\sqrt{4aL}$ Thus tension at the left end of moving part is $\frac{\lambda v^2}{2} = 2\lambda aL$ and

now the pseudo force is λLa thus net force applied by the hand must be $2\lambda aL + \lambda aL = 3\lambda aL$ Ans.

Think: How will you find Heat lost and work done in the process?

Solution: Heat liberated is frame independent. From Hand frame entire pseudo potential energy is lost. Thus heat lost is $\lambda L \times aL$ ($\because mg$ Ans) i.e $\Delta Q = \lambda L^2 a$ Ans. Now final $KE = \frac{1}{2}\lambda L v^2 = \frac{1}{2}\lambda L \times 4aL = 2\lambda aL^2$
Thus net work done: $\Delta W = \Delta Q + \Delta KE = 3\lambda L^2 a$ Ans.

EXAMPLE: PATHFINDER: MCQ 20:



Water is flowing through two rigidly connected coplanar U tubes. In the left tube of cross-sectional area A, speed of water flow is u and in the right tube of cross-sectional area $A/2$ speed water flow is unknown. If the net force on the tube assembly due to water flow is zero, what must be the speed of the water flow in right tube? Neglect effect of gravity.

(a) $\frac{u}{2}$ (b) $\frac{u}{\sqrt{2}}$

(c) $u\sqrt{2}$ (d) $2u$

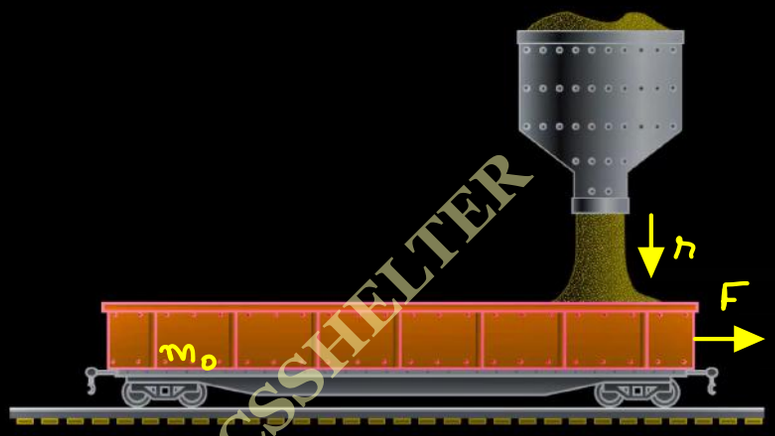
Solution: Thrust force on water in left tube: $F_L = \rho A u (2u)$
$\Rightarrow F_L = 2\rho A u^2$ similarly if v is the speed in right tube, thrust force $F_R = \left(\rho \frac{A}{2} v\right)(2v) = \rho A v^2$. Now the tubes

experience the reactions of these thrust forces. Thus for equilibrium, $F_R = F_L \Rightarrow 2u^2 = v^2 \Rightarrow v = \sqrt{2}u$ Ans (option c).

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Example: Pathfinder: Build: 36:

When an empty freight train of mass m_0 starts, loading of coal in the train begins at a constant rate r from a stationary hopper. If the track is horizontal and engine pull F is a constant, deduce expression for speed of the train as a function of time t. Neglect all resistive forces.



Solution: By impulse momentum theorem, $\int \vec{F} dt = \Delta \vec{p}$. Now, mass at a general time t is $(m_0 + rt)$. If v is the velocity by this time, By Impulse Momentum Theorem:

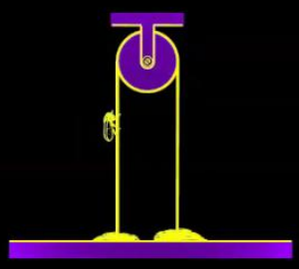
$$Ft = (m_0 + rt)v$$

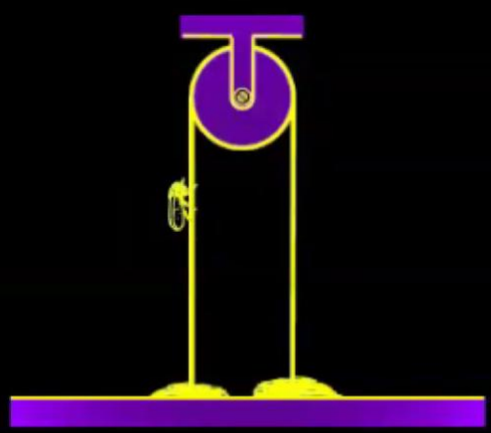
$$\Rightarrow v = \frac{Ft}{(m_0 + rt)} \quad \text{Ans.}$$

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Example: Pathfinder: Build 37:

A uniform thread of linear mass density $\lambda = 2.0$ g/m passes through a fixed ideal pulley. At its ends, the thread is stacked into two heaps that do not interfere motion of the thread. A beetle of mass $m = 80$ g tries to manage itself at a constant height on the thread by adjusting its speed relative to the thread. At some steady speed of the thread, the beetle succeeds in doing so. Find this steady speed of the thread. Acceleration due to gravity is $g = 10$ m/s^2.



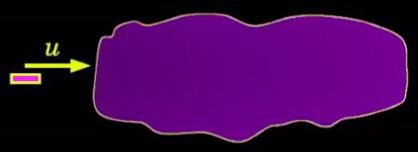


Solution: This question is very similar to a previous question where chain was moving from one pile to another lower pile. Here since Beetle manages to stay at its place, it must be pushing the chain down by a force equal to its weight. Thus, in steady state this pushing force on LHS must be balanced by pulling force of tension on the other side.

Thus $mg = \lambda v^2 \Rightarrow v = \sqrt{\frac{mg}{\lambda}} = 20 \text{ ms}^{-1}$ Ans.

EXAMPLE: PATHFINDER: CHECK-24:

A spaceship of mass m with its engine off moving in free space with a velocity u enters a cloud of uniform density ρ and length l. Particles forming the cloud strike the spaceship and stick to it. How much time will the spaceship take to cross the cloud? For simplicity, assume the spaceship as a cylinder of circular section area S.



Solution: When the spaceship has travelled a distance x, by momentum conservation, its speed is given by the equation:

$$mu = (m + \rho S x)\, v \quad \text{thus:}$$

$$v = \left(\frac{m}{m + \rho S x}\right) u = \frac{dx}{dt} \Rightarrow \int_0^l (m + \rho S x)\, dx = \int_0^T mu\, dt$$

$$\Rightarrow \quad muT = ml + \frac{\rho S l^2}{2} \Rightarrow T = \frac{l}{u} + \frac{\rho S l^2}{2mu} = \frac{l}{u}\left[1 + \frac{\rho S l}{2m}\right] \text{ Ans.}$$

EXAMPLE: PATHFINDER: CHECK:25:

In rocket propulsion, kinetic energy imparted to the burnt fuel as ejected gases is wasted as far as propelling is concerned. If at an instant v is the speed of the rocket and u is the speed of the ejected gases relative to the rocket, develop suitable expression for efficiency of propulsion.

Solution: Thrust force on the rocket: $F_x = u\, dm/dt$

$\Rightarrow$ Power developed on the rocket $u\frac{dm}{dt} \times v = P$

KE of exhausted gases from space frame:

$$dKE = \frac{1}{2} dm\,(u - v)^2 \Rightarrow \frac{dKE}{dt} = \frac{1}{2}\frac{dm}{dt}(u - v)^2 = \dot{K}$$

Now efficiency of propulsion: $\eta = \dfrac{P}{P + \dot{K}}$

$$\Rightarrow \eta = \frac{uv}{uv + \frac{1}{2}(u - v)^2} = \frac{2uv}{u^2 + v^2} \quad \text{Ans.}$$



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